Chapter 5 • Linear Equations and Inequalities

Solved Exercise 5.1

1. Solve the six equations.

(i) $12x=-36$
$\Rightarrow x=-3$
(ii) $\frac{x}{3}=-18$
$\Rightarrow x=-54$
(iii) $\frac{-x}{4}=\frac{1}{12}$
$\Rightarrow x=\frac{-1}{3}$
(iv) $2=26x+28$
$\Rightarrow x=-1.$
(v) $4(2x-1)-9x=10$
$\Rightarrow -x-4=10$
$\Rightarrow x=-14.$
(vi) $\frac{-x}{2}=9-2x$
$\Rightarrow \frac{3x}{2}=9$
$\Rightarrow x=6.$

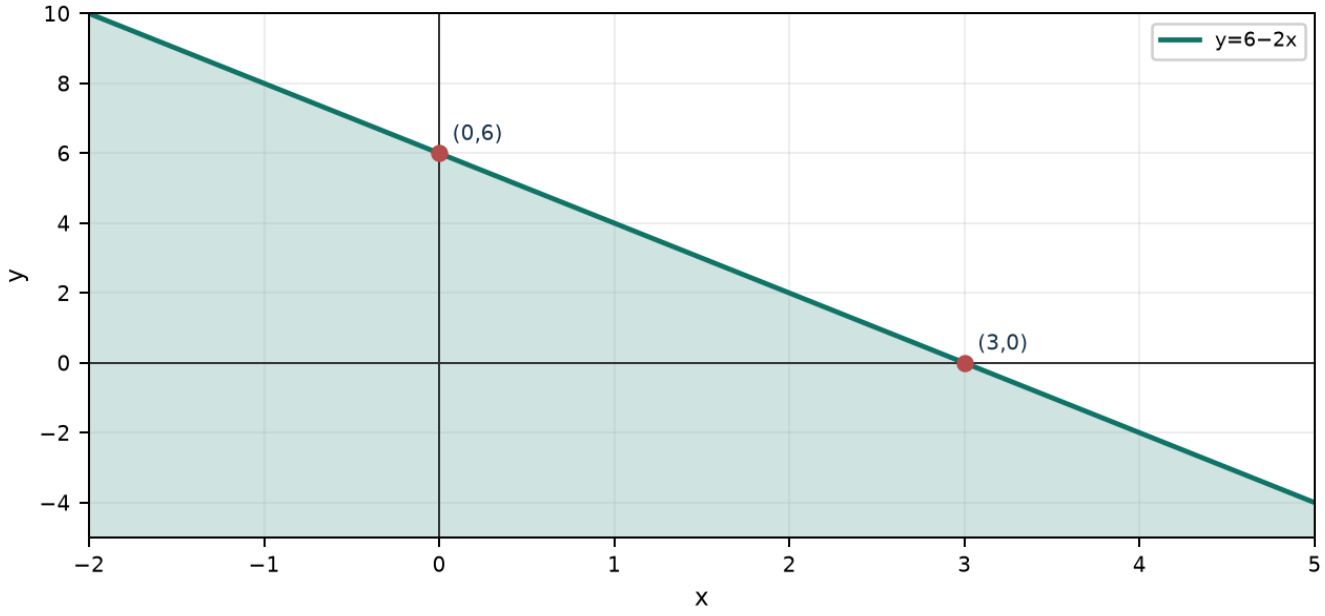
2. Solve each inequality.

(i) $x\leq 4$
$(-\infty,4]$
(ii) $7>x$, hence $x<7$
$(-\infty,7)$
(iii) $2x\geq 0$, hence $x\geq 0.$
$[0,\infty)$
(iv) $x+10\leq 0$
$\Rightarrow x\leq -10.$
$(-\infty,-10]$
(v) $20x-9<-1$
$\Rightarrow 20x<8$
$\Rightarrow x<\frac{2}{5}$
$(-\infty,\frac{2}{5})$
(vi) $x-2\leq -4+2x \Rightarrow 2\leq x$, hence $x\geq 2.$
$[2,\infty).$

x	boundary $y=6-2x$
-1	8
0	6
3	0
4	-2

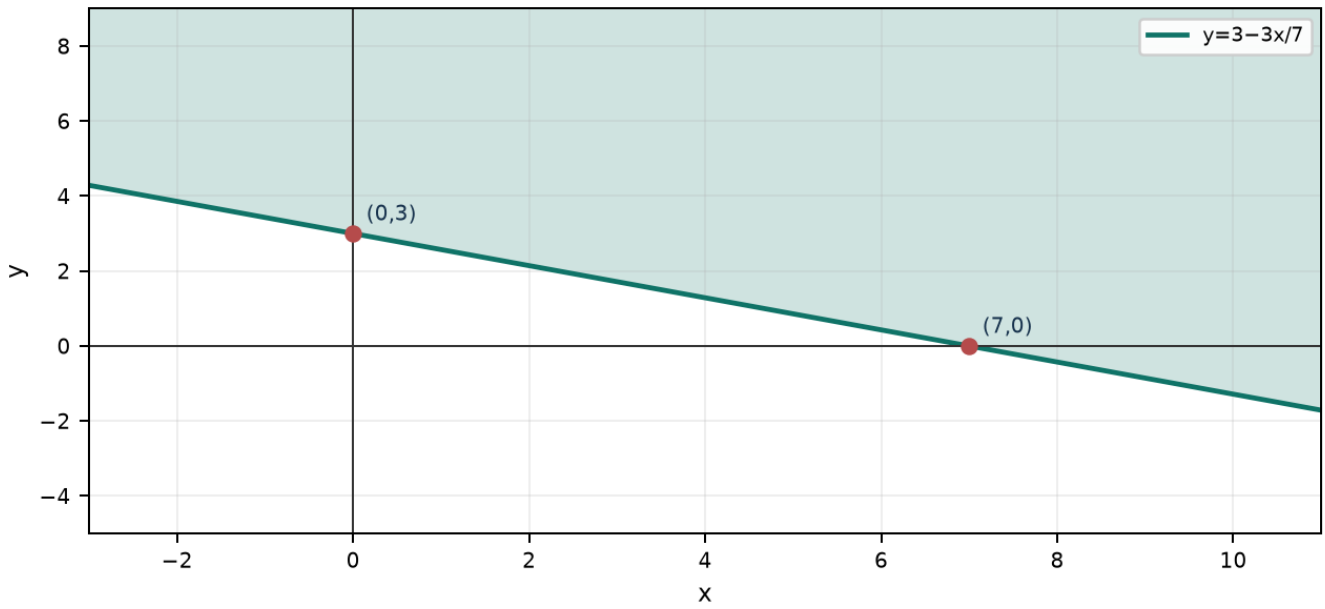
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Exercise 5.1 Q3(i): y ≤ 6 − 2x



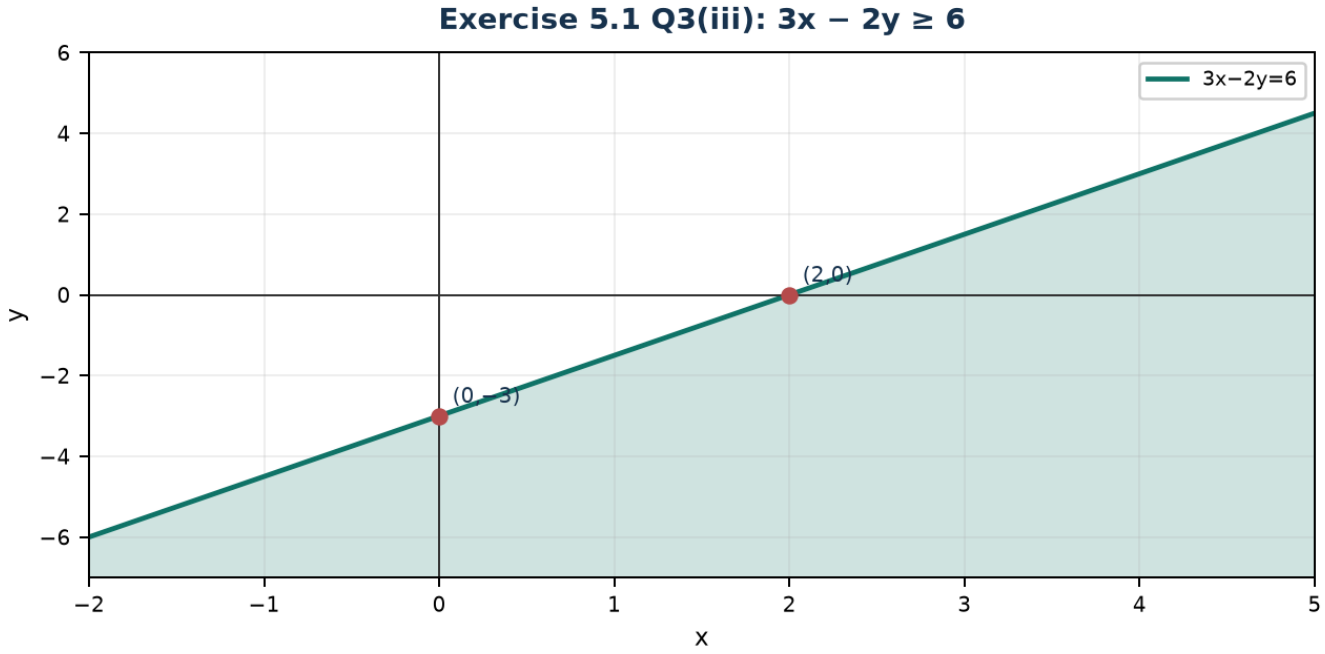
x	boundary y=3−3x/7
-7	6
0	3
7	0
14	-3

Exercise 5.1 Q3(ii): y ≥ 3 − 3x/7

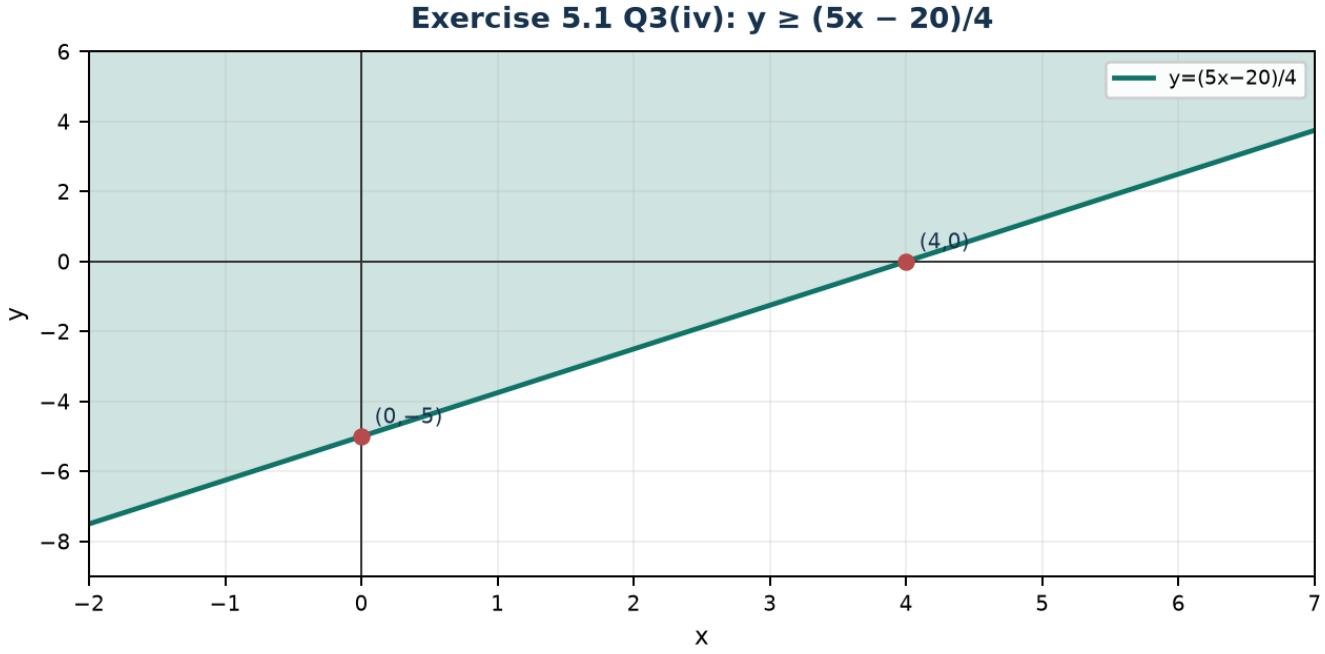


x	boundary y=(3x−6)/2
-2	-6
0	-3

x	boundary y=(3x−6)/2
2	0
4	3

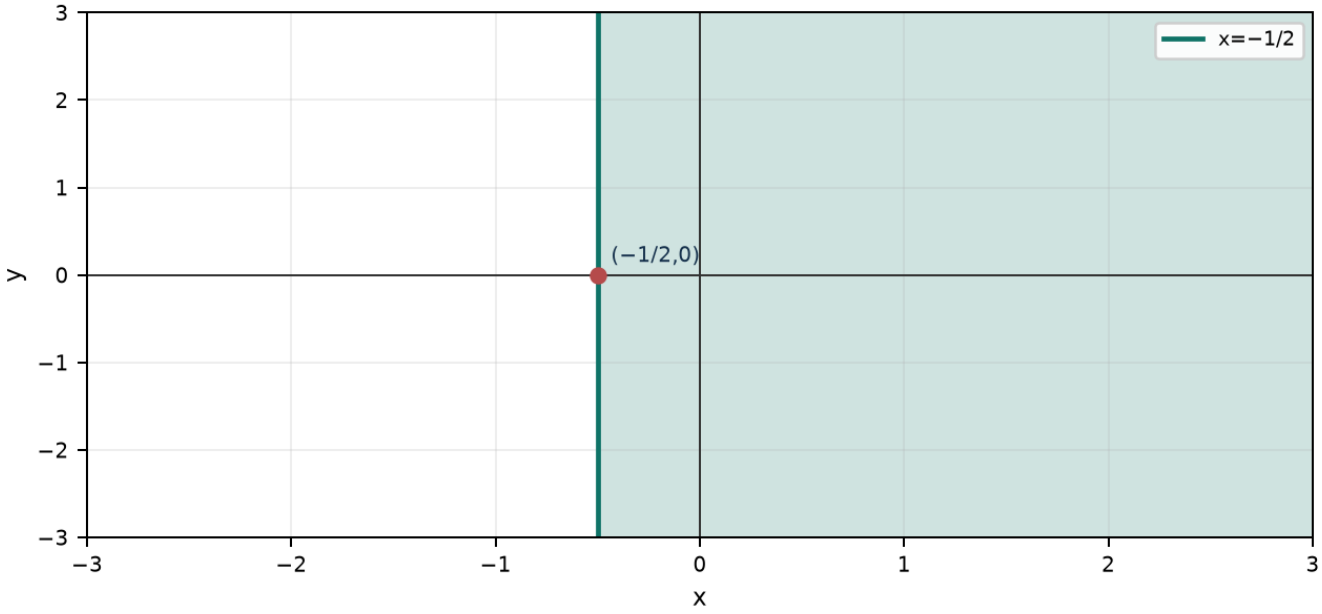


x	boundary y=(5x−20)/4
0	-5
2	-2.5
4	0
6	2.5



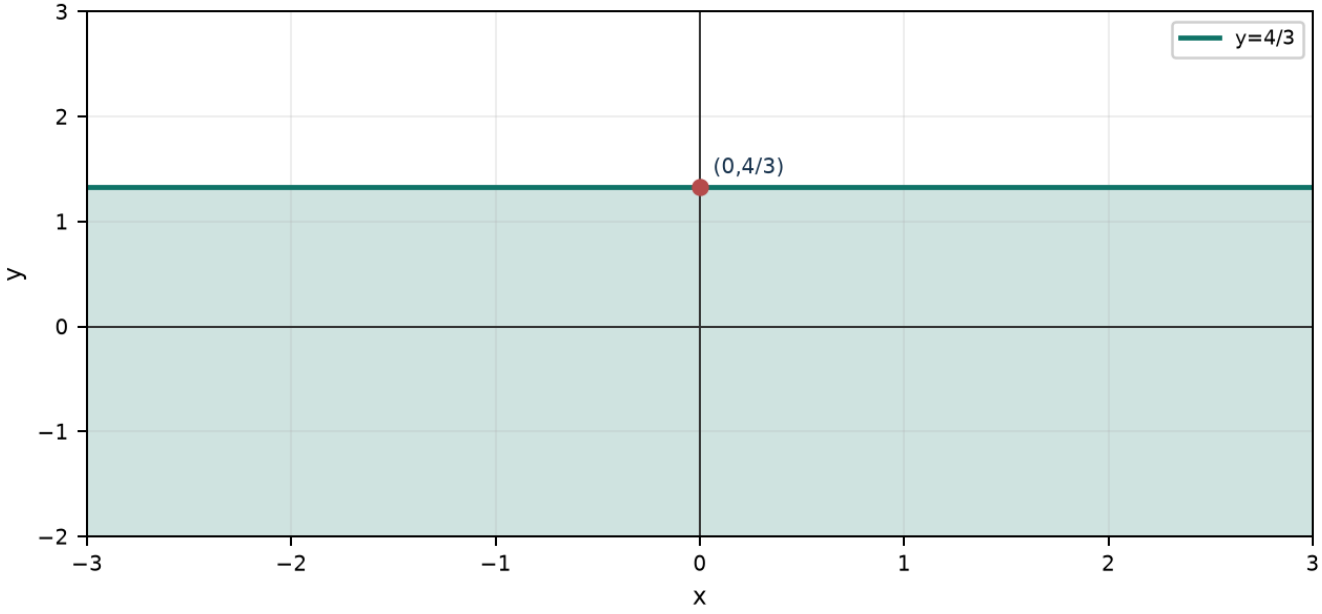
boundary x	y
−1/2	−2
−1/2	0
−1/2	2

Exercise 5.1 Q3(v): x ≥ −1/2



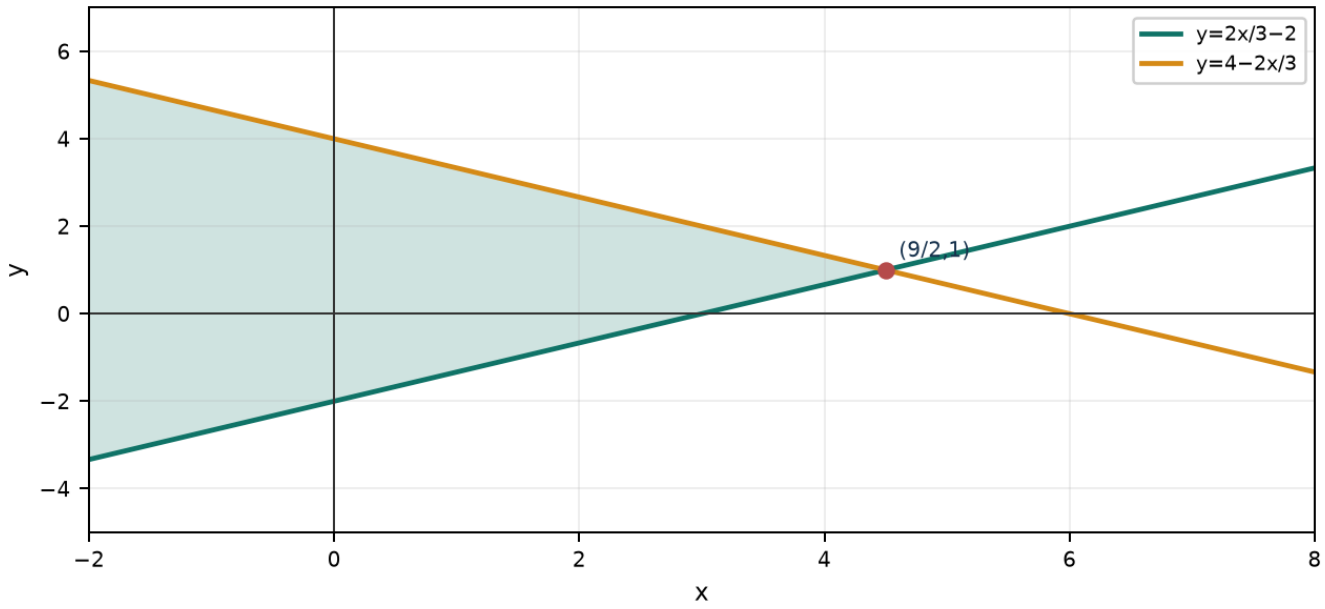
x	boundary y
−2	4/3
0	4/3
2	4/3

Exercise 5.1 Q3(vi): y ≤ 4/3



x	y=2x/3−2	y=4−2x/3
0	-2	4
3	0	2
4.5	1	1
6	2	0

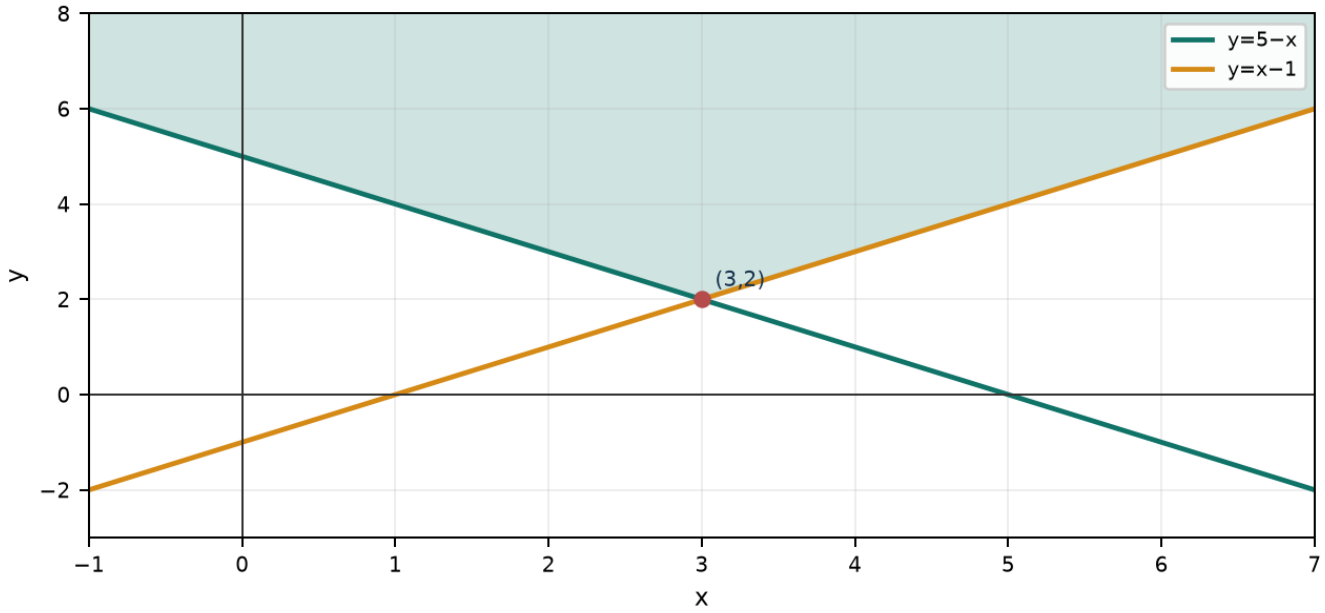
Exercise 5.1 Q4(i): simultaneous solution region



x	y=5−x	y=x−1
0	5	-1
1	4	0
3	2	2
5	0	4

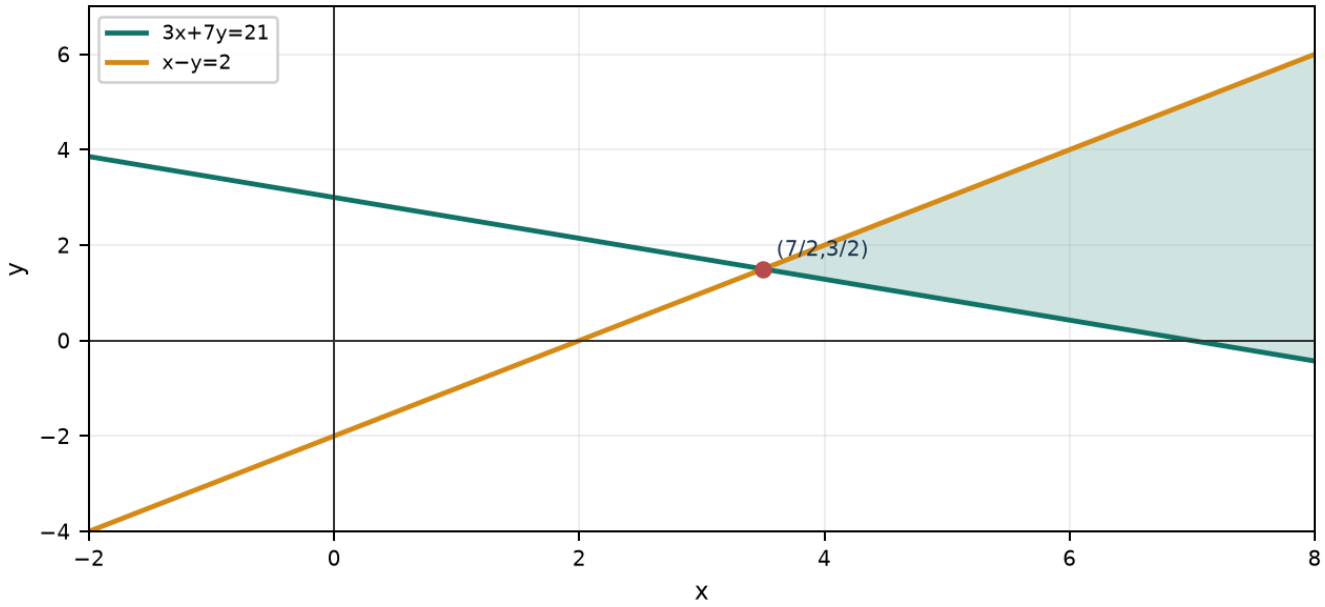
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Exercise 5.1 Q4(ii): y≥5−x and y≥x−1



x	y=3−3x/7	y=x−2
0	3	-2
2	2.1429	0
3.5	1.5	1.5
7	0	5

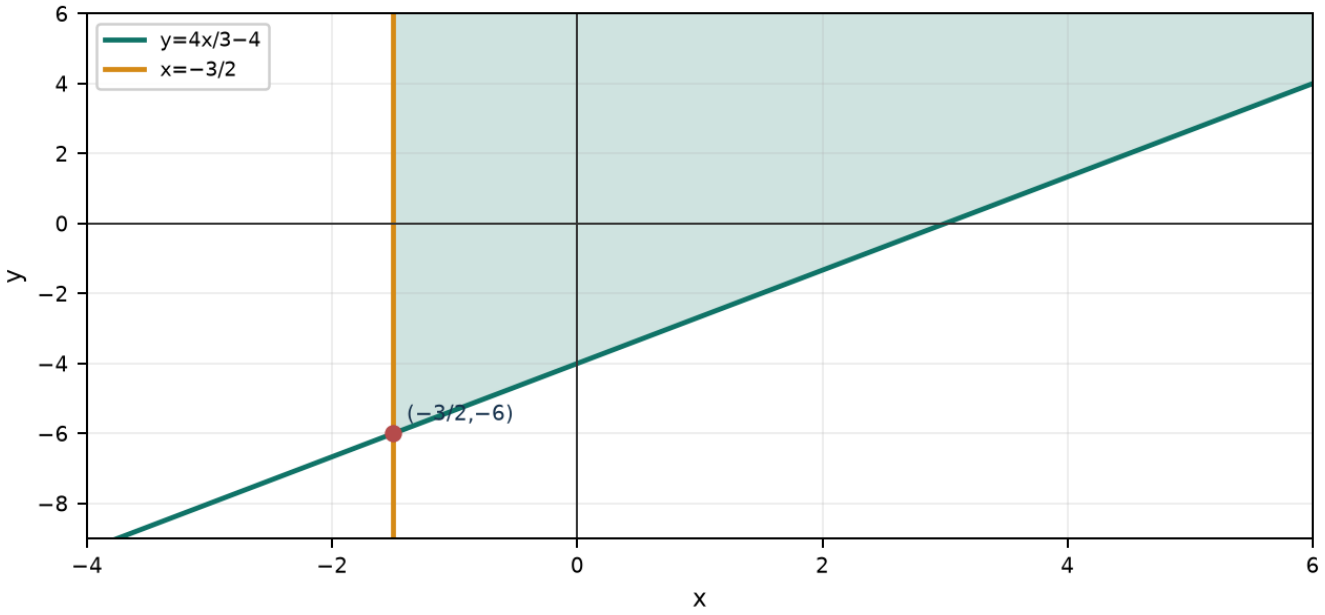
Exercise 5.1 Q4(iii): 3x+7y≥21 and x−y≥2



x	boundary y=4x/3−4
-1.5	-6
0	-4

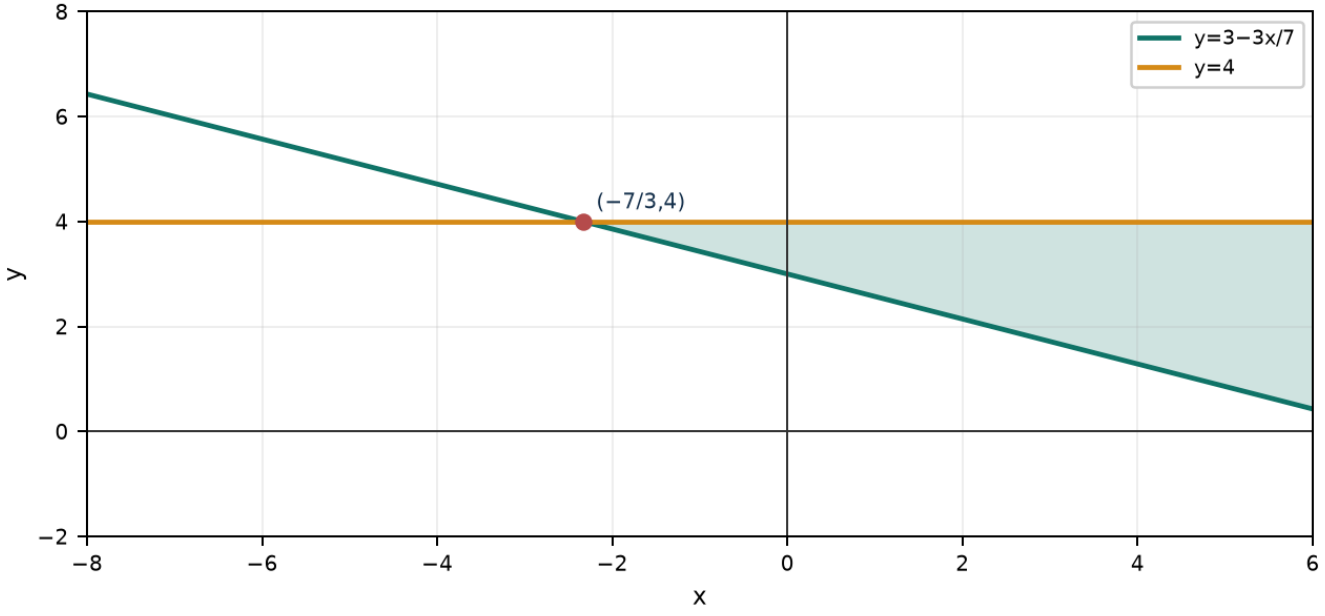
x	boundary y=4x/3−4
3	0
6	4

Exercise 5.1 Q4(iv): sloped and vertical boundaries

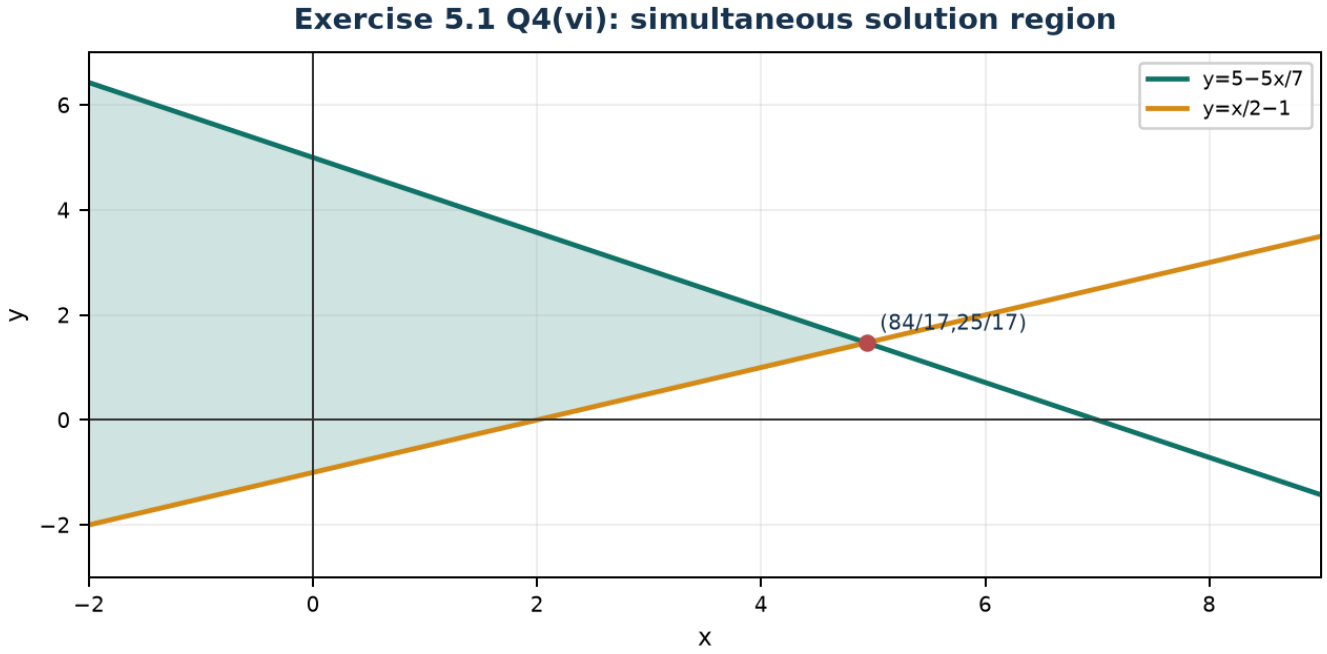


x	y=3−3x/7	y=4
-7	6	4
-2.3333	4	4
0	3	4
7	0	4

Exercise 5.1 Q4(v): sloped and horizontal boundaries



x	y=5−5x/7	y=x/2−1
0	5	-1
2	3.5714	0
4.9412	1.4706	1.4706
7	0	2.5



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